LEARNER GUIDE

The angle at the centre

Circle Theorems · Lesson 2. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

Take two points **A** and **B** on a circle. Any angle standing on the arc they cut off **subtends** it. The **angle at the centre**, ∠AOB, is made by the two radii; an **angle at the circumference**, ∠APB, is made by two chords from a point P on the far arc. This lesson's keystone theorem: **the angle at the centre is exactly twice the angle at the circumference standing on the same arc** — so ∠AOB = 2 × ∠APB. Read backwards, the circumference angle is half the centre angle. A striking consequence: every point P on that arc gives the *same* ∠APB, so all such angles are equal. The reason is a single extra line: join P to O and extend it, and two **isosceles** triangles (equal radii) do the rest.

What you are learning to notice

- Angle at the **centre** = 2 × angle at the **circumference**, on the **same arc**.
- Read both ways: given the centre, **halve**; given the circumference, **double**.
- Every angle at the circumference on the same arc is **equal** (all are half of ∠AOB).
- Always give the theorem as your **reason** — a circle-theorem answer needs a justification, not just a number.

Moving through it, one step at a time

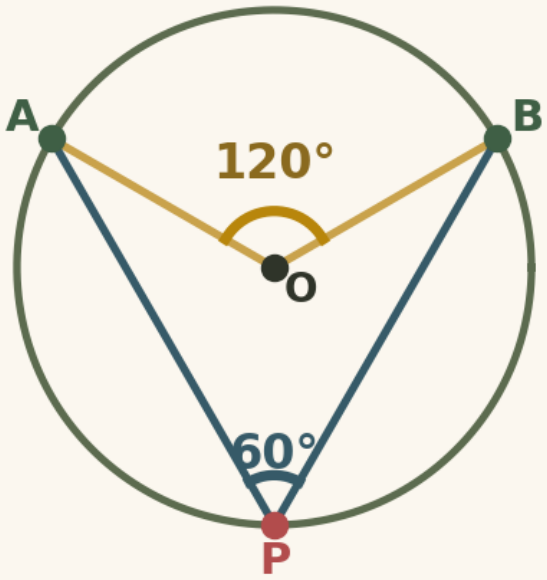
Identify the arc both angles stand on. Spot which angle you are given: if it is at the centre, halve it; if it is at the circumference, double it. Write the result and name the theorem as the reason. For the reflex case (P on the minor arc), the centre angle is the reflex one, and it is still twice the circumference angle.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Use the theorem both ways

O is the centre. The angle at the centre ∠AOB is 120°. Find the angle at the circumference ∠APB.

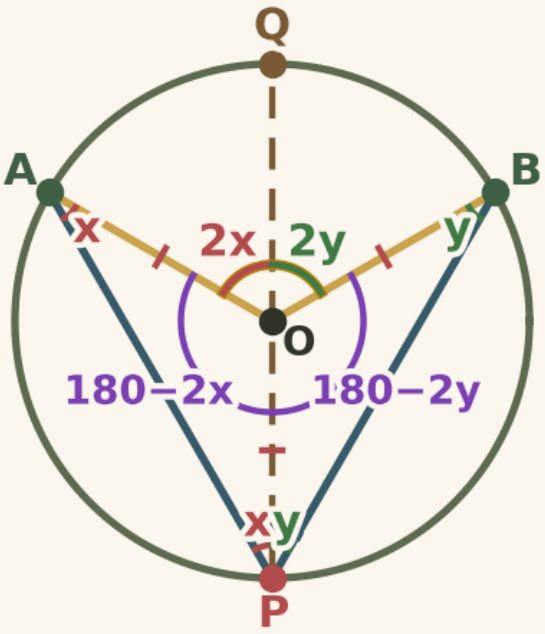


1. Both angles stand on the same arc AB.	∠AOB and ∠APB on arc AB
2. Angle at the centre = 2 × angle at the circumference.	120 = 2 × ∠APB
3. So halve the centre angle.	∠APB = 120 ÷ 2

∠APB = 60°, because the angle at the centre is twice the angle at the circumference on the same arc.

Example 2 — Why it is exactly double

Prove ∠AOB = 2 × ∠APB. Join P to O and extend it to Q.



1. OA = OB = OP are radii, so triangles OAP and OBP are isosceles.	equal radii
2. Isosceles ⇒ equal base angles: ∠OAP = ∠OPA = x and ∠OBP = ∠OPB = y.	∠APB = $x + y$
3. Angles in each triangle sum to 180°, so the apex angles are ∠AOP = 180 – 2x and ∠BOP = 180 – 2y.	apex angles
4. P, O, Q lie on a straight line, so ∠AOQ = 180 – (180 – 2x) = 2x, and ∠BOQ = 2y.	∠AOQ = 2x, ∠BOQ = 2y

∠AOB = 2x + 2y = 2(x + y) = 2 × ∠APB. The angle at the centre is twice the angle at the circumference. ■

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the parts of a circle and 'standing on an arc' (Lesson 1), angles in a triangle sum to 180°, and isosceles base angles are equal), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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